

AP Chem Test 1

30 September 2026

Atomic structure and periodicity

Average atomic mass

Average atomic mass is the summation of each mass number multiplied by its relative abundance (%).

Example: Neon has three isotopes: Ne-20 (19.992 amu, 90.48%), Ne-21 (20.994 amu, 0.27%), and Ne-22 (21.991 amu, 9.25%).

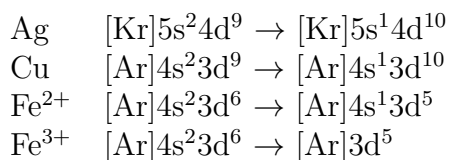
To find the average atomic mass:

$$19.992 \times 0.9048 + 20.994 \times 0.0027 + 21.991 \times 0.0925 = \boxed{20.180 \text{ amu}}$$

Electron configuration and d-orbital stabilisation

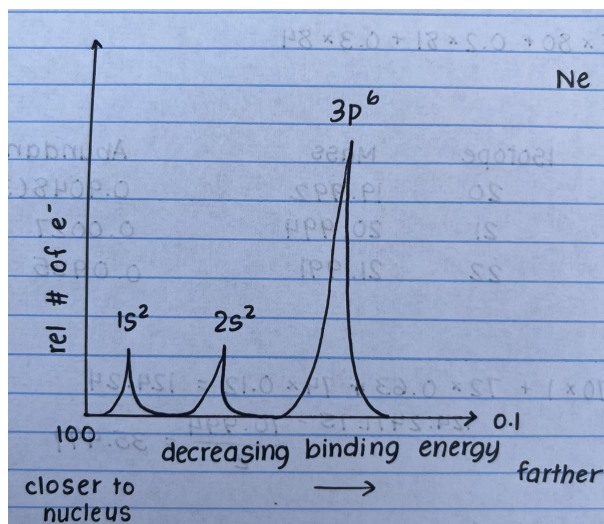
d-orbital stabilisation refers to the d-orbitals preferring to be half-filled or completely filled (5 or 10) rather than slightly underfilled or overfilled (4, 9).

The noble gas configuration of chromium, for example, appears to be $[\text{Ar}]4s^23d^4$. However, to stabilise the d-orbital, one electron is borrowed from the s-orbital, making the final noble gas configuration $[\text{Ar}]4s^13d^5$. Similarly:



For ions, electrons are removed from the outermost level (4 for the two iron ions shown).

Photo electron spectroscopy



Periodic table trends and Coulomb's Law

Coulomb's Law states that

$$F \propto \frac{q_1 q_2}{r^2} \quad (1)$$

where F is the force (repulsive if positive, attractive if negative), q_1 and q_2 are the respective charges, and r is the distance between the two particles.

This becomes important when considering periodic table trends such as:

ionisation energy (IE) energy required to remove an electron from a gaseous atom or ion

electron affinity (EA) energy released with the addition of an electron to a gaseous ion or atom

electronegativity (EN) an atom's tendency to attract electrons toward itself

EA and size increase down and to the left.

In a given period, the greater the number of protons, the smaller the atom will be. However, as you go down the periodic table, the number of electron shells increases along with size. Electron affinity increases down and to the left, but its absolute value (since more energy released = more negative) increases up and to the right.

IE and EN increase up and to the right.

Ionisation energy increases as you go up, since the number of electron shells decreases, making the radius (r^2) smaller. Across the period, the number of protons increases, pulling the electron shell closer to it, making it harder to remove an electron. The higher positive charge pulling the electrons close to the nucleus is also the reason for the trend in electronegativity.

Multiple ionisation energies

The first, second, third, and fourth ionisation energies of Al are as follows:

$$I_1 = 580 \text{ kJ/mol}$$

$$I_2 = 1815 \text{ kJ/mol}$$

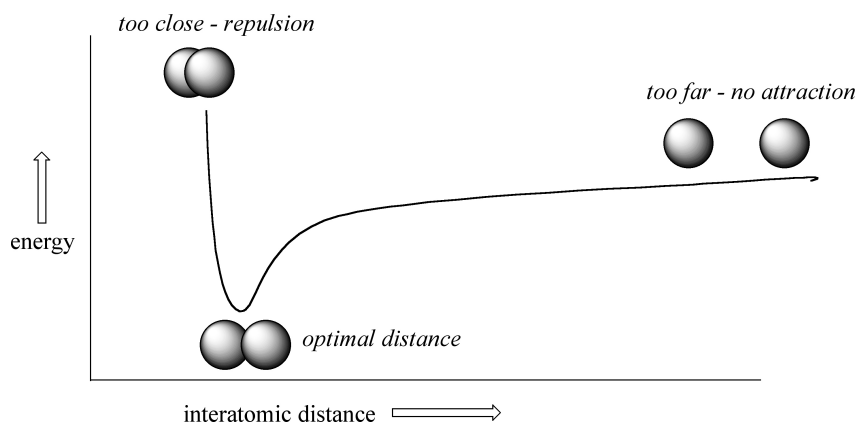
$$I_3 = 2740 \text{ kJ/mol}$$

$$I_4 = 11600 \text{ kJ/mol}$$

The jump in ionisation energy from I_3 to I_4 can be attributed to Al's position on the periodic table; as it loses three electrons, it is stripped of its outer electron shell, giving it a full octet and making it harder to remove another electron from the new valence shell.

Bonding

Bond length diagram



The optimal distance (most stable) depends on the radius of the atom. If the atom is bigger, the optimal distance will be longer (graph shifts to the right). The **average bond length** is found at the bottom of this curve.

Energy effects in ionic compounds

Lattice energy is the change in energy that takes place when separated gaseous ions pack together to form an ionic solid.

$$\text{Lattice energy} \propto \frac{q_1 q_2}{r}$$

Between NaCl and KCl, Na and K have the same charge, but NaCl has the shorter internuclear distance (Na is smaller than K), so there is more energy released.

Between NaF and MgF₂, Mg is slightly smaller than Na, but it has a higher charge, so there is more energy released in MgF₂.

Hybridisation

sigma (σ) bond bond on axis between two nuclei (2 electrons)

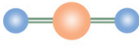
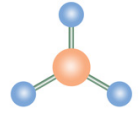
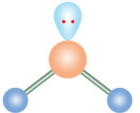
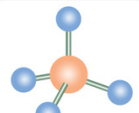
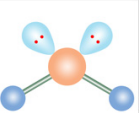
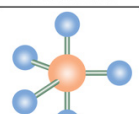
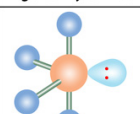
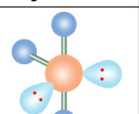
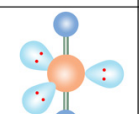
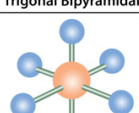
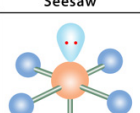
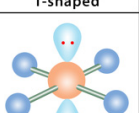
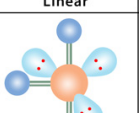
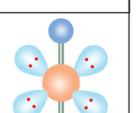
pi (π) bond bond formed around sigma bond by 2 p-orbitals (2 electrons)

Each electron domain can have at most one sigma bond. All others are pi bonds.

Electron and molecular geometries

Electron geometries are named by the first row in the table below and are determined by the number of electron domains only.

VSEPR Theory Chart

Number of Electron Groups	Lone Pairs = 0	Lone Pairs = 1	Lone Pairs = 2	Lone Pairs = 3	Lone Pairs = 4
2	 Linear				
3	 Trigonal Planar	 Angular or Bent			
4	 Tetrahedral	 Trigonal Pyramidal	 Angular or Bent		
5	 Trigonal Bipyramidal	 Seesaw	 T-shaped	 Linear	
6	 Octahedral	 Square Pyramidal	 Square Planar	 T-shaped	 Linear

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N.B. Row 3 and below can expand their octets. HONC = 1234 (hydrogen prefers 1 bond, oxygen 2, nitrogen 3, and carbon 4) and remembering to neutralise formal charge, especially on the central atom, are useful for drawing. The equation for formal charge is as follows:

$$\text{formal charge} = \text{valence electrons} - \text{bonds} - \text{individual lone pair electrons} \quad (2)$$

(For example, an oxygen atom with 4 lone pair electrons and 2 bonds has a formal charge of 0. An oxygen atom with 6 lone pair electrons and 1 bond has a formal charge of -1.)

Intermolecular forces

Types of IMFs, in order of strength (weakest → strongest)

London Dispersion Forces (LDFs)

- In all molecules
- Depends on **instantaneous dipoles** formed by random movement of electrons and induced dipoles, a temporary separation of positive and negative
- The bigger the molecule, the greater the LDFs

Dipole-dipole (dip-dip)

Between **two polar molecules only**.

Hydrogen bonds (H-bonds)

- Stronger version of dipole-dipole
- Attraction of H in H–O, H–F, H–N to an electronegative nucleus (at least 1 molecule needs to have this bond)

Ion-dipole

- Between an **ionic compound** and a **polar molecule**
- **In order for an ionic compound to be soluble, ion-dipole force must overcome IMFs of solvent and ionic attraction**

Physical effects of IMFs

IMFs influence solubility, melting point, boiling point, viscosity, and surface tension.

Solubility depends on whether the IMFs between the solute particles and the solvent particles overcome the IMFs between the solute particles and themselves, as well as the solvent particles and themselves.

Boiling point and melting point depend on the strength of the IMFs. If the IMFs are stronger, the boiling and melting points will both be elevated.

Spectroscopy

The higher the energy, the shorter the wavelength, the deeper the penetration is into the molecule.

From low energy to high energy:

Form of radiation	Effects
microwave	molecular rotation
infrared	molecular vibration (bond stretch)
visible and ultraviolet	transfer of valence electrons
X-ray	transfer of core electrons
gamma rays	nuclear effects

IMF-based separation techniques

Chromatography separates components of the analyte based on compounds' relative attraction to the stationary phase vs. the mobile phase.

There are three parts to chromatography—the **mobile phase**, **stationary phase**, and the **analyte**. The stationary phase is the fixed medium (e.g. paper), the mobile phase is the moving fluid (e.g. acetone), and the **analyte** is the sample moving along the stationary phase (e.g. ink).

How much an analyte moves depends on the IMFs formed with both the paper and the solvent. If the analyte is insoluble in the solvent, it will not move; the more soluble it is, the higher it will move.

Distillation separates compounds based on boiling points.

Properties of solutions

Vapour pressure is the pressure above a solution or a liquid exerted by the solution or liquid. When **vapour pressure = atmospheric pressure**, the solution has reached its boiling point. Higher IMFs lead to a lower vapor pressure.

When a solute is added to a compound, it causes **boiling point elevation** and **freezing point depression**. This is because the solute disrupts the formation of the lattice structure, lowering the freezing point, and adding a solute to a liquid lowers its vapour pressure, increasing its boiling point.

Deviation from Ideal Gas Law behaviours

The particles of ideal gasses have **no volume** and **no IMFs**. As the volume of the particles and the IMFs increase, the deviation increases.

At very high pressures, this deviation becomes significant.

Gas Laws:

$$\begin{aligned}\text{pressure} &\propto \frac{1}{\text{volume}} \\ \text{pressure} &\propto \text{temperature} \\ \text{volume} &\propto \text{temperature}\end{aligned}$$

Kinetic molecular theory of gasses

$$KE = \frac{1}{2}mv^2 \quad (3)$$

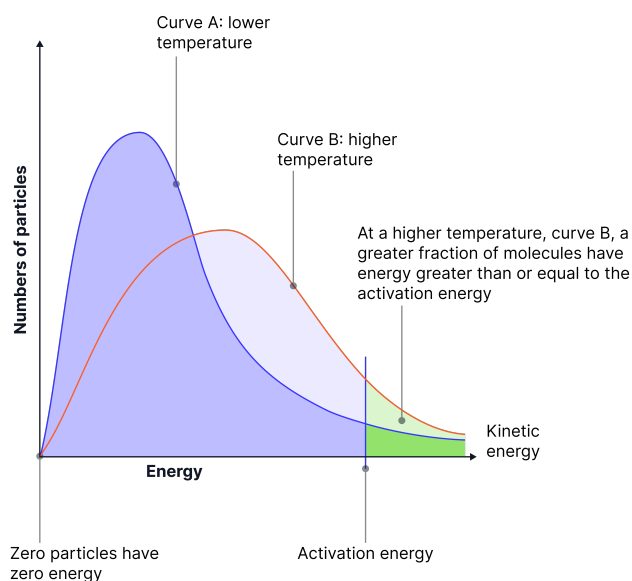
At pressure = 1 atm, we can postulate:

1. Particles are so far apart that the volume is negligible
2. Particles are in constant motion, and pressure is a result of collisions with the container
3. Particles do not exert force on each other (no IMFs)
4. Average KE $\propto$ temp in K

Every gas has the same average KE at a given temperature but different velocities.

$$V \propto \sqrt{\text{mass}}$$

Boltzmann distribution



At the highest point of each curve is the **average velocity**. As the mass increases, the average velocity decreases.

Empirical and molecular formulae

The empirical formula is the ratio between molecules in a compound.

Example: A compound is 43.45% phosphorus, 56.36% oxygen, and has a mass of 283.88 g/mol. What are its empirical and molecular formulae?

Assuming 100g:

$$43.46 \text{ g} \cdot \frac{1 \text{ mol}}{30.96 \text{ g}} = 1.403 \text{ mol P}$$

$$56.36 \text{ g} \cdot \frac{1 \text{ mol}}{16 \text{ g}} = 3.5225 \text{ mol O}$$

$$1.403/1.403 = 1 \text{ mol P}, 3.5225/1.403 = 2.5 \text{ mol O}$$

Multiply by 2 to eliminate the decimal—2 mol P, 5 mol O

P_2O_5 is the empirical formula.

To find the molecular formula, use the molar mass:

$$\text{Molar mass of } \text{P}_2\text{O}_5 = 141.94 \text{ g/mol}$$

$$283.88/141.94 \approx 2$$

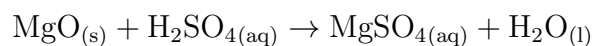
P_4O_{10} is the molecular formula.

Percent yield

% of theoretical yield produced in a lab setting

$$\text{percent yield} = \frac{\text{theoretical yield}}{\text{experimental yield}} \times 100\% \quad (4)$$

Example: What is the max mass of $\text{MgSO}_4 \cdot 7 \text{H}_2\text{O}$ formed if 100g MgCO_3 is heated? If only 268g was produced, what was the percent yield?



($\text{MgSO}_{4(aq)} + \text{H}_2\text{O}_{(l)}$ becomes $\text{MgSO}_4 \cdot 7 \text{H}_2\text{O}$)